Ninad Rajendra Rade

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EDUCATION

New York University, Tandon School of Engineering — New York, NY
M.S. Computer Engineering — GPA: 3.84 / 4.0 — Aug 2025 – May 2027

University of Mumbai – Fr. C. Rodrigues Institute of Technology — Mumbai, India
B.E. Computer Engineering, Honors in AI & ML — CGPA: 8.97 / 10 — Jul 2021 – Jul 2025

Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Probability & Statistics, Linear Algebra, Convex Optimization

TECHNICAL SKILLS

Languages: Python, SQL, C/C++, JavaScript, Bash
ML / AI Frameworks: PyTorch, TensorFlow, Scikit-learn, XGBoost, Hugging Face Transformers, Random Forest, SMOTE
LLM / GenAI: LLM API Integration, Prompt Engineering, Retrieval-Augmented Generation (RAG), SHAP, Explainable AI (XAI)
CV / NLP / Data: OpenCV, MediaPipe, Pose Estimation, NLTK, TF-IDF, LSTM, PySpark, Spark SQL, Spark MLlib
Tools & Platforms: AWS S3, REST APIs, Streamlit, MySQL, MongoDB, Git, Linux, Pandas, NumPy, Matplotlib

EXPERIENCE

DavaNinja Pvt. Ltd. — Mumbai, India
Software Engineering Intern – Healthcare AI Systems — Oct 2024 – Mar 2025

- Engineered Python and NLP preprocessing pipelines for 100K+ clinical pharmacy records, producing clean structured inputs for drug-classification and risk-assessment ML models in a healthcare AI system.
- Implemented schema validation and statistical drift checks, cutting manual review effort by **40–50%** and improving downstream model reliability.
- Integrated REST APIs across 3 internal services to automate data ingestion, eliminating a 2-week bottleneck and accelerating classification workflow delivery.
- Authored and executed **40+ manual test cases** for pre-release mobile builds, cutting QA review cycles by **30%**.

SELECTED PROJECTS

DeepFit – Real-Time Pose Estimation & Form Correction | *Python, OpenCV, MediaPipe, Scikit-learn, Computer Vision*

- Developed a real-time CV pipeline with OpenCV and MediaPipe Pose to extract 3D anatomical landmarks for live exercise form tracking at **30+ FPS** via multi-threaded CPU inference.
- Converted raw landmark coordinates into joint-angle features using vector dot products; trained a Random Forest ensemble achieving **94% F1-score** classifying correct form vs. momentum-compensated reps, with per-rep error tracking and UI alerts.

ScentGraph – Explainable ML Longevity Predictor | *Python, XGBoost, TF-IDF, SHAP, Streamlit, Explainable AI*

- Designed a tabular ML pipeline that maps olfactory ingredient profiles and concentration tiers to predicted fragrance longevity and projection scores.
- Applied TF-IDF vectorization across 200+ aromatic compounds and fine-tuned an XGBoost Regressor, reducing longevity prediction error to under **0.8 MAE hours**.
- Integrated SHAP explainability into a Streamlit dashboard for custom profile input, instant inference, and localized feature importance breakdowns.

NYC Taxi Demand Prediction & Anomaly Detection | *Python, PySpark, Spark MLlib, Streamlit*

- Engineered a distributed PySpark pipeline processing 8.8M cleaned trips from ∼10M raw records in under 3 minutes via caching and broadcast joins; Random Forest achieved **RMSE 41.35, R^2 0.69** – a 4× improvement over baseline.
- Shipped a Streamlit dashboard with a hybrid IQR + rule-based anomaly detector for interactive temporal and spatial demand exploration.

Stock Market Prediction with LSTM | *Python, TensorFlow, NumPy, Pandas, Time-Series Forecasting*

- Constructed an LSTM forecasting pipeline with rolling-window feature engineering and chronological train/test splits; reduced prediction error by **18%** over baseline via MAE/RMSE-guided hyperparameter tuning.
- Deployed an interactive web app interface for live predictions, enabling real-time inference and visualization of forecasted price trends.

CERTIFICATIONS

MLOps Certification – Duke University *(In Progress)* | **IBM AI Engineering Professional Certificate** – IBM *(In Progress)*
Generative AI Fundamentals – Databricks, 2026 | **Software Engineer** – HackerRank, 2026

LEADERSHIP & SOCIAL IMPACT

AIESEC in India — Mumbai, India
Junior Executive → OC Vice President, Partnerships & Logistics — May 2022 – Aug 2022

- Led operations and partnerships for youth development programs with **50+ participants**; authored Python automation scripts to standardize reporting workflows and improve data accuracy for non-technical stakeholders.

Other Roles: Sponsorship Head – Computer Society of India | Team Member – National Service Scheme | Editor – Manthan Book Club | Member – Rotary Club